

```

from pyspark.sql import SparkSession
from pyspark.sql.functions import col

spark = SparkSession.builder \
    .appName("Week6 Assignment") \
    .getOrCreate()

d:\Samradni Dahiphale\Projects\Celebal task\
Data_Engineering_Internship_Celebal_Technologies\.venv\Lib\site-
packages\pyspark\testing\utils.py:127: FutureWarning: PySpark does not
yet fully support pandas >= 3.0.0. Some features may not work
correctly. It is recommended to use pandas < 3.0.0 for now.
    require_minimum_pandas_version()

df = spark.read \
    .option("header", True) \
    .option("inferSchema", True) \
    .csv("data/source.csv")

print("Original Data")

df.show()

Original Data
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+
|product_id|  category|price|stock|old_name|base_price|user_id|
|status|amount|region|priority|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+
|      101|Electronics|55000|  15| Laptop|      55000|  1001|
Completed|  1200| North|   High|
|      102| Furniture|12000|  20|  Chair|      12000|  1002|
Pending|   800| South|   Low|
|      103|Electronics|75000|  10| Mobile|      75000|  NULL|
Completed|  2500| East| Medium|
|      104| Clothing| 2500|  40| Tshirt|       2500|  1004|
Completed|   500| North|   High|
|      105|Electronics|15000|  18| Speaker|      15000|  1005|
Pending|  1800| West|   High|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+

print("Schema")

df.printSchema()

Schema
root
|-- product_id: integer (nullable = true)

```

```
|-- category: string (nullable = true)
|-- price: integer (nullable = true)
|-- stock: integer (nullable = true)
|-- old_name: string (nullable = true)
|-- base_price: integer (nullable = true)
|-- user_id: integer (nullable = true)
|-- status: string (nullable = true)
|-- amount: integer (nullable = true)
|-- region: string (nullable = true)
|-- priority: string (nullable = true)
```

```
print("Select Product ID and Price")
```

```
df.select("product_id", "price").show()
```

```
Select Product ID and Price
```

```
+-----+-----+
|product_id|price|
+-----+-----+
|      101|55000|
|      102|12000|
|      103|75000|
|      104| 2500|
|      105|15000|
+-----+-----+
```

```
#Filter Electronics
```

```
print("Electronics Products")
```

```
df.filter(df.category=="Electronics").show()
```

```
Electronics Products
```

```
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+
|product_id|category|price|stock|old_name|base_price|user_id|
status|amount|region|priority|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+
|      101|Electronics|55000| 15| Laptop| 55000| 1001|
Completed| 1200| North| High|
|      103|Electronics|75000| 10| Mobile| 75000| NULL|
Completed| 2500| East| Medium|
|      105|Electronics|15000| 18| Speaker| 15000| 1005|
Pending| 1800| West| High|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+
```

```
#Rename Column
```

```
df = df.withColumnRenamed("old_name","new_name")
```

```
print("Renamed Column")
```

```
df.show()
```

```
Renamed Column
```

```
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+
|product_id|  category|price|stock|new_name|base_price|user_id|
status|amount|region|priority|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+
|      101|Electronics|55000|  15|  Laptop|      55000|  1001|
Completed|  1200| North|   High|
|      102| Furniture|12000|  20|   Chair|      12000|  1002|
Pending|   800| South|   Low|
|      103|Electronics|75000|  10|  Mobile|      75000|  NULL|
Completed|  2500| East|  Medium|
|      104| Clothing| 2500|  40|  Tshirt|       2500|  1004|
Completed|   500| North|   High|
|      105|Electronics|15000|  18| Speaker|      15000|  1005|
Pending|  1800| West|   High|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+
```

```
#Cast Price
```

```
df = df.withColumn(
    "price",
    col("price").cast("double")
)
```

```
print("Price Converted")
```

```
df.printSchema()
```

```
Price Converted
```

```
root
```

```
|-- product_id: integer (nullable = true)
|-- category: string (nullable = true)
|-- price: double (nullable = true)
|-- stock: integer (nullable = true)
|-- new_name: string (nullable = true)
|-- base_price: integer (nullable = true)
|-- user_id: integer (nullable = true)
|-- status: string (nullable = true)
|-- amount: integer (nullable = true)
|-- region: string (nullable = true)
```

```

|-- priority: string (nullable = true)

#Add New Column
df = df.withColumn(
    "final_price",
    col("base_price")*1.18
)

print("Final Price")

df.show()

Final Price
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
|product_id| category| price|stock|new_name|base_price|user_id|
status|amount|region|priority|final_price|
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
|      101|Electronics|55000.0| 15| Laptop| 55000| 1001|
Completed| 1200| North| High| 64900.0|
|      102| Furniture|12000.0| 20| Chair| 12000| 1002|
Pending| 800| South| Low| 14160.0|
|      103|Electronics|75000.0| 10| Mobile| 75000| NULL|
Completed| 2500| East| Medium| 88500.0|
|      104| Clothing| 2500.0| 40| Tshirt| 2500| 1004|
Completed| 500| North| High| 2950.0|
|      105|Electronics|15000.0| 18| Speaker| 15000| 1005|
Pending| 1800| West| High| 17700.0|
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+

#Filter Orders
print("Completed Orders")

df.filter(
    (col("status")=="Completed") &
    (col("amount")>1000)
).show()

Completed Orders
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
|product_id| category| price|stock|new_name|base_price|user_id|
status|amount|region|priority|final_price|
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
|      101|Electronics|55000.0| 15| Laptop| 55000| 1001|
Completed| 1200| North| High| 64900.0|

```

103	Electronics	75000.0	10	Mobile	75000	NULL
Completed	2500	East	Medium	88500.0		

#Handle Null Values

`print("Non Null User IDs")`

```
df.filter(
    col("user_id").isNotNull()
).show()
```

Non Null User IDs

product_id	category	price	stock	new_name	base_price	user_id
status	amount	region	priority	final_price		
101	Electronics	55000.0	15	Laptop	55000	1001
Completed	1200	North	High	64900.0		
102	Furniture	12000.0	20	Chair	12000	1002
Pending	800	South	Low	14160.0		
104	Clothing	2500.0	40	Tshirt	2500	1004
Completed	500	North	High	2950.0		
105	Electronics	15000.0	18	Speaker	15000	1005
Pending	1800	West	High	17700.0		

#Filter OR Condition

`print("North OR High")`

```
df.filter(
    (col("region")== "North") |
    (col("priority")== "High")
).show()
```

North OR High

product_id	category	price	stock	new_name	base_price	user_id
status	amount	region	priority	final_price		
101	Electronics	55000.0	15	Laptop	55000	1001
Completed	1200	North	High	64900.0		
104	Clothing	2500.0	40	Tshirt	2500	1004
Completed	500	North	High	2950.0		

105	Electronics	15000.0	18	Speaker	15000	1005
Pending	1800	West	High	17700.0		

#Wide Transformation

```
print("Group By Category")
```

```
df.groupBy("category").count().show()
```

Group By Category

category	count
Electronics	3
Clothing	1
Furniture	1

#Save as CSV

```
df.write \
.mode("overwrite") \
.option("header", True) \
.csv("output/csv_output")
```

#Save as Parquet

```
df.write \
.mode("overwrite") \
.parquet("output/parquet_output")
```

#Read Parquet

```
parquet_df = spark.read.parquet("output/parquet_output")
```

```
print("Reading Parquet")
```

```
parquet_df.show()
```

Reading Parquet

product_id	category	price	stock	new_name	base_price	user_id
status	amount	region	priority	final_price		
101	Electronics	55000.0	15	Laptop	55000	1001
Completed	1200	North	High	64900.0		
102	Furniture	12000.0	20	Chair	12000	1002
Pending	800	South	Low	14160.0		
103	Electronics	75000.0	10	Mobile	75000	NULL

```
Completed| 2500| East| Medium| 88500.0|
| 104| Clothing| 2500.0| 40| Tshirt| 2500| 1004|
Completed| 500| North| High| 2950.0|
| 105| Electronics| 15000.0| 18| Speaker| 15000| 1005|
Pending| 1800| West| High| 17700.0|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
```

```
#Stop Spark
spark.stop()
```